Grading form - Assignment 676932 - Student 513818

• First score each criterion according to the tutorial instructions below.
• Then sum points separately for **#R**, **#Radv**, and **#Python**, and then scale to grades:
--> R = summed score / 47 * 8
--> Radv = summed score / 3 * 2
--> Python = summed score / 10 * 2
--> Total = R + max(Radv, Python)
• Your final grade should use **#R** plus the higher of **#Python** vs **#Radv**.
• Your report includes a preliminary AI estimate which you can use to guide your self-grading.
• Pay special attention to the consistency histogram as inconsistent LLM scoring may indicate unreliable AI feedback.

Dont forget to compute final self scoring grades below!

Self scoring:
• #R: 8
• #Radv: 2
• #Python:

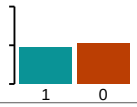
--> Overall: 10
Based on **#R** and

Preliminary AI estimate:
• #R: **6.81**
• #Radv: **2.0**
• #Python: **0.0**

--> Overall: **8.81**
Based on **#R** and **#Radv**.

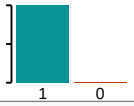
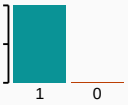
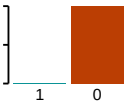
Tutorial example

Below is an example of a single question table.

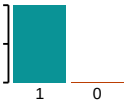
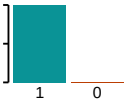
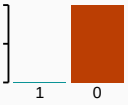
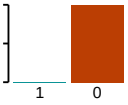
Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (21 regrades)
Correctly explained what `getwd()` does.		0	The AI estimated that this criterion was not met.	

• Open your PDF in a PDF viewer that allows you to add comments.
• For each criterion, check if your code / answer meets it and award yourself 0 or 1 point.
• After you are done, upload your annotated PDF as a comment to your assignment submission on Canvas.
• We will review your self-grading and adjust it if necessary.

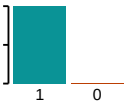
Question: #R1

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. The dataset is loaded from the specified GitHub URL		1	The dataset is loaded using read.csv from the specified GitHub URL, matching rubric criterion 1.	
2. The dataset is used for a scatterplot		1	The plot() function is used on two numeric vectors from the dataset, producing a scatterplot as required by rubric criterion 2.	
3. DATE is mapped to the x-axis of the scatterplot	1	0	The student maps airport$year to the x-axis instead of DATE, so DATE is not mapped as required by rubric criterion 3.	
4. TAVG is mapped to the y-axis of the scatterplot		1	The student maps airport$TAVG to the y-axis of the scatterplot, satisfying rubric criterion 4.	

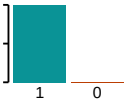
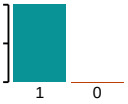
Question: #R2

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. ggplot is used to create a bar plot of the titanic_train data		1	ggplot is used with geom_bar() and titanic_train as the data source, satisfying the rubric requirement for a bar plot of titanic_train.	
2. Sex is mapped to the x-axis (must be visually on the x-axis; so, if axes are flipped, y-axis is acceptable)		1	Sex is mapped to aes(x = Sex) in ggplot, so Sex is visually on the x-axis as required by the rubric.	
3. Survived is mapped to the fill aesthetic		1	Survived from titanic_train is converted to the survival factor and used as fill in aes(fill = survival), satisfying the rubric's requirement that Survived be mapped to the fill aesthetic.	
4. Legend labels for fill are "dead" and " alive" (labels must match exactly)	1	0	The factor labels are set to 'Dead' and ' Alive', which differ in capitalization from the exact required labels 'dead' and 'alive' in the rubric.	
5. The legend title for fill is "How did it go?" (title must match exactly)	1	0	The legend title is set with labs(fill = ' Did they make it?'), which does not match the exact required title 'How did it go?' in the rubric.	

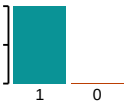
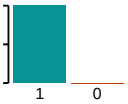
Question: #R3

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. A theme (e.g., theme_bw()) is explicitly mentioned		1	The student explicitly mentions several ggplot themes (theme_light, theme_dark, theme_minimal), satisfying the rubric requirement of explicitly naming a theme.	
2. An explanation of the theme choice is provided (can be very brief)		1	The student explains their preference for theme_minimal in one sentence, stating it removes extra lines and makes the plot look simple/clean, which fulfills the rubric's explanation requirement.	

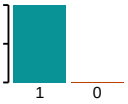
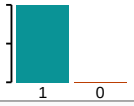
Question: #R4

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. The improved plot shows the relationship between cars$speed and cars$dist		1	The rubric requires the improved plot to show the relationship between cars$speed and cars$dist; the student uses plot(cars$speed, cars$dist) so the relationship is correctly displayed.	
2. A first improvement is listed AND implemented (any improvement that improves understanding or visual clarity is valid as long as it is implemented and listed; e.g., adding title, labels, proper axes ranges, ticks, ...)		1	The rubric requires a first improvement to be listed and implemented; adding a title ('main = "Car Distance vs Speed"') is implemented and listed as '#Added a title'.	
3. A second improvement is listed AND implemented (any improvement that improves understanding or visual clarity is valid as long as it is implemented and listed; e.g., adding title, labels, proper axes ranges, ticks, ...; must be different from the first improvement)		1	The rubric requires a second, distinct improvement to be listed and implemented; adding x and y labels ('xlab', 'ylab') is implemented and listed as '# Added x and y labels'.	
4. A third improvement is listed AND implemented (any improvement that improves understanding or visual clarity is valid as long as it is implemented and listed; e.g., adding title, labels, proper axes ranges, ticks, ...; must be different from the first two improvements)		1	The rubric requires a third, distinct improvement to be listed and implemented; changing point style via ' pch = 20' is implemented and listed as ' ##Changed point style'.	

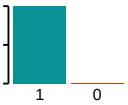
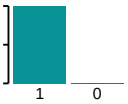
Question: #R5

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. ggplot is used to create a bar plot showing the relationship between chickens and maximum weight		1	The student uses ggplot() with geom_col() to create a bar plot showing Chick (chickens) versus weight, which represents maximum weight per chicken as required by the rubric.	
2. Only chickens 1, 20, 5, 40, and 19 are included in the plot		1	The data is subset to fivechick using ChickWeight$Chick %in% c(1, 20, 5, 40, 19), so only chickens 1, 20, 5, 40, and 19 are included in the plot as specified by the rubric.	
3. Chickens are shown on the x-axis AND follow the order: 1, 20, 5, 40, 19		1	Chick is converted to a factor with levels = c(1, 20, 5, 40, 19), and this factor is mapped to the x-axis in ggplot, ensuring chickens are on the x-axis in the correct order per the rubric.	
4. The maximum weight is computed correctly (using any valid method) AND is shown on the y-axis		1	The maximum weight per Chick is computed with aggregate(weight ~ Chick, fivechick, max), and the resulting weight variable is mapped to the y-axis in ggplot, satisfying the rubric requirement for correct max weight computation and display.	

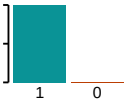
Question: #R6

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. ggplot is used to create a scatterplot with the cars dataset (e.g., geom_point() or other solutions resulting in a scatterplot)		1	Meets rubric: ggplot() is used with data = cars and geom_point() to produce a scatterplot using the cars dataset.	
2. speed is shown on the x-axis		1	Meets rubric: speed is mapped to the x aesthetic in mapping = aes(speed, dist).	
3. dist is shown on the y-axis		1	Meets rubric: dist is mapped to the y aesthetic in mapping = aes(speed, dist).	
4. A correct regression line is added (e.g., geom_smooth(aes(speed, dist), method = 'loess', formula = 'y ~ x') or any solution resulting in exactly the same regression line)		1	geom_smooth() is added without changing method or formula, which uses the same data and aesthetics as the points, resulting in the default loess curve equivalent to the reference regression line described in the rubric.	

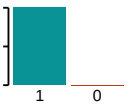
Question: #R7

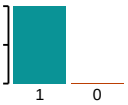
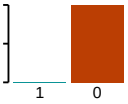
Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. ggstatsplot is used to create a plot with the ToothGrowth dataset (any plot, created with the package and using the dataset is acceptable)		1	The code uses the ggstatsplot package and explicitly sets data = ToothGrowth, which matches the rubric requirement that any plot created with ggstatsplot using the ToothGrowth dataset is acceptable.	
2. The plot shows the relationship between supp and len		1	The plot is created with ggbetweenstats using x = supp and y = len, directly showing the relationship between supplement type (supp) and tooth length (len) as required by the rubric.	
3. An APPROPRIATE ggstatsplot is created (any plot that answers the research question; e.g., ggbetweenstats() or similar)		1	The function ggbetweenstats is a between-subject comparison appropriate for answering whether tooth growth differs between orange juice and meds, satisfying the rubric's requirement for an appropriate ggstatsplot (e.g., ggbetweenstats).	

Question: #R8

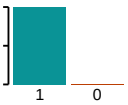
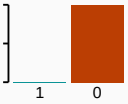
Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. plotly is used to create a 3D scatter plot that includes TotalHeight, HeadCircumference, and ShoulderToWaist (all three variables must be included but can be mapped to any of the three axes)		1	The student uses plot_ly from plotly and includes TotalHeight, HeadCircumference, and ShoulderToWaist as x, y, and z aesthetics, creating a 3D scatter plot; this matches the rubric requirement that all three variables be included on any axes.	
2. All axes are labeled using the tilde notation (e.g., ~TotalHeight) or another method that results in correct axis labels		1	All three aesthetics x, y, and z are specified with tilde notation (~TotalHeight, ~ShoulderToWaist, ~HeadCircumference), which according to the rubric results in correct axis labels.	

Question: #R9

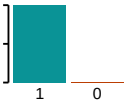
Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. The cran_downloads() function is used to get the data AND a line plot of CRAN download data is created (any line plot is acceptable, even if it does not match the reference plot exactly)		1	The student uses cran_downloads() to get data and creates a line plot of CRAN download data with geom_line(), satisfying the rubric requirement for a line plot of download data.	

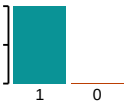
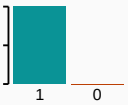
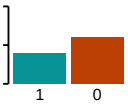
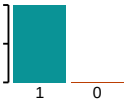
Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
2. date, count, and package are correctly mapped to x, y, and color aesthetics respectively		1	In ggplot(downs, aes(date, count, color = package)), date is mapped to x, count to y, and package to color exactly as specified in the rubric.	
3. The plot only shows the "ggplot2" and "plotly" packages		1	cran_downloads(packages = c("ggplot2", "plotly"), ...) restricts the data to only these two packages, so the plot shows only "ggplot2" and " plotly" as required by the rubric.	
4. The y-axis label is "Package Downloads", and the title is "Package popularity over time" (label and title must match exactly)	1— made the gif file as required by the question with x and y labels. Exact replication was not specified	0	The student does not add labs() or otherwise set the y-axis label to " Package Downloads" or the title to " Package popularity over time", so the exact label and title specified in the rubric are missing.	
5. The plot is rendered or saved as a GIF (any method that would render the plot inside the IDE or save it as a GIF file is acceptable)		1	The plot is animated with animate() and saved as a GIF using anim_save("cran_downloads.gif", animation), which meets the rubric requirement to render or save the plot as a GIF.	

Question: #R10

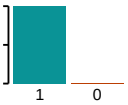
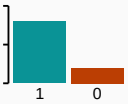
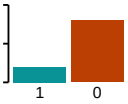
Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. A first valid style improvement is described (e.g., using informative variable names, breaking up code into multiple lines, using informative comments, etc.)		1	The student describes improving the clarity of plotting code by indenting, breaking to new lines, and using consistent spacing, which the rubric lists as valid style improvements such as breaking up code into multiple lines and improving readability.	
2. A second valid style improvement is described (e.g., using informative variable names, breaking up code into multiple lines, using informative comments, etc.; must be different from the first improvement)	1- improved style of two questions as required by the question, distinct edits was not specified	0	The second description again focuses on clarity of plotting code by breaking to a new line after each argument and aligning indents; this is another instance of breaking up code into multiple lines/spacing and is not clearly a different type of improvement as required by the rubric.	

Question: #R11

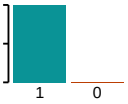
Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. Correct explanation of what the code does (e.g., "function v returns TRUE if variable b and function input x have the same number of characters", or similar correct explanation; can be very brief)		1	Student states the function checks if the lengths of two strings are equal after splitting into letters, which matches the rubric description of v returning TRUE if b and x have the same number of characters.	

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
2. The student identified a first poor style aspect (e.g., uninformative variable names, inconsistent assignment operators, mixing global and local variables in function, improper indentation, lack of comments, redundant code, one-line if statement, inconsistent spacing, [there might be others])		1	Student notes 'poorly defined function name', which is a poor style aspect related to uninformative naming, matching examples in the rubric.	
3. The student identified a second poor style aspect (e.g., uninformative variable names, inconsistent assignment operators, mixing global and local variables in function, improper indentation, lack of comments, redundant code, one-line if statement, inconsistent spacing, [there might be others]; must be different from the first aspect)		1	Student comments 'vague variable name assignment' for yy, identifying uninformative variable naming as a second distinct poor style aspect.	
4. The student identified a third poor style aspect (e.g., uninformative variable names, inconsistent assignment operators, mixing global and local variables in function, improper indentation, lack of comments, redundant code, one-line if statement, inconsistent spacing, [there might be others]; must be different from the first two aspects)	1-previous was function name, this was variable name— distinct poor styling choices	0	Student cites 'unclear variable naming' for lxx as another instance of uninformative naming, but this is not a different aspect from previous naming issues required by the rubric.	
5. The student identified a fourth poor style aspect (e.g., uninformative variable names, inconsistent assignment operators, mixing global and local variables in function, improper indentation, lack of comments, redundant code, one-line if statement, inconsistent spacing, [there might be others]; must be different from the first three aspects)		1	Student identifies 'reused function name for variable', which is a distinct poor style aspect involving mixing function and variable names, different from earlier naming issues.	
6. The student identified a fifth poor style aspect (e.g., uninformative variable names, inconsistent assignment operators, mixing global and local variables in function, improper indentation, lack of comments, redundant code, one-line if statement, inconsistent spacing, [there might be others]; must be different from the first four aspects)		1	Student mentions lack of indentation ('should be indented for readability'), which is a distinct poor style aspect of improper indentation per the rubric.	

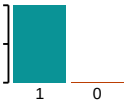
Question: #R12

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. A 3×3 matrix is created		1	Rubric 1 requires a 3×3 matrix; the student uses matrix(1:9, 3, 3, byrow = TRUE), which constructs a 3×3 matrix (dim = c(3,3)).	
2. ONLY a single function is used to create a 3x3 matrix (ignore print() or similar functions that just display the matrix)		1	Rubric 2 requires only a single function to create the 3×3 matrix; the submission uses a single call to matrix() (multiplication by 1:3 is an operator, not another function).	
3. ONLY a single function is used to create a 3x3 matrix AND it contains the correct values (row 1: 1, 2, 3; row 2: 8, 10, 12; row 3: 21, 24, 27) (one valid solution is `matrix(1:9, 3, byrow = TRUE) * 1:3`, but there might be others)	1 it does match	0	Rubric 3 requires the matrix to have specific values: rows (1,2,3), (8,10,12), (21,24,27). The student's code matrix(1:9, 3, 3, byrow = TRUE) * 1:3 produces rows (1,2,3), (4,10,18), (7,16,27), which do not match the required values.	

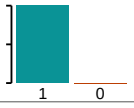
Question: #R13

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. Any shortcut is mentioned in a comment (the student may have changed the keybinding so any shortcut is acceptable)		1	The student mentions a shortcut (Ctrl+Shift+A) inside an R comment (##control shift A), satisfying the rubric requirement that any shortcut be mentioned in a comment.	

Question: #R14

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. Code is used to create plot or image WITHOUT loading any local files		1	The student uses R code with the memer package to generate a meme image directly, without loading any local files, satisfying rubric criterion 1.	
2. The plot or image qualifies as a meme (i.e., it is humorous or satirical in nature)		1	The generated image uses a known meme template (Y-U-NOguy) with humorous text about making memes in R before Friday night plans, so it qualifies as a meme per rubric criterion 2.	

Question: #Radv1

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
1. Random art is created (anything qualifies as art)		1	The code uses ggplot to draw a polygon with randomized radii, which qualifies as random art per criterion 1.	

Rubrics	Self scoring (enter 0 or 1)	AI score (indicator)	AI 'thought'	AI Consistency (5 regrades)
2. set.seed() is used and affects the generated art		1	set.seed() is called before runif(), which controls the random radii and thus affects the generated art, satisfying criterion 2.	
3. Only base R or ggplot2 are used		1	The solution uses only base R functions (set.seed, seq, runif, cos, sin, sample, data.frame) and ggplot2, meeting criterion 3.	

Dont forget to compute final self scoring grades and enter them on the first page!